Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			The {vibrations / oscillations} (1)... ...are {at right angles / at 90° / perpendicular } to the <u>direction</u> of {energy transfer /wave travel / propagation / motion} (1)	2			2		
	(b)	(i)		1.2[2] in middle column (2 or 3 sig figs) (1) 5.85 in final column (3 sig figs) (1) accept 5.86		2		2	2	
		(ii)		When depth changes from 1.0 to 4.0 (quadrupled) (1), the speed doubles from 3.13 to 6.26 (pair of values quoted) (1) Accept other pairs of values, e.g. 0.5 m $\rightarrow$ 2.0 m (1) and calculated speed for 2. 0 m (4.426) using equation compared with speed for 0.5 m (2.21) (1) To award 2 marks 'statement is true' [or equiv] must be made NB purely mathematic argument based upon $\sqrt{4} = 2 \rightarrow$ 0 marks			2	2	2	
		(iii)		Depth of water and speed scales chosen to accommodate more than half the graph (1) (Best scales are 0.5 m per square and 1 m/s per square) All 6 <u>given</u> points plotted within tolerance of <1 small square [ignore origin] (2) 5 <u>given</u> points correctly plotted within tolerance of <1 small square (1) Smooth curve drawn through the points from d = 1 m to d = 4 m within <1 small square tolerance with attempt to draw curve to origin (1) NB. Use of given data values to label major lines on speed axis $\rightarrow$ 0 marks Missing 2 m on the depth axis $\rightarrow$ lose the scales mark only and curve needs to be best fit. If vertical scale is for $\sqrt{d}$ $\rightarrow$ 0 marks	1	2 1		4	4	

	(c)	(i)		Use of value taken from graph within < 0.1 at 2.0 m [expect 4.4 m/s] (1) Substitution: wavelength = $\frac{4.4}{0.2}$ (1) = 22.0 m (1) Use of $v = 3.13\sqrt{d}$ to give 4.43 m/s $\rightarrow$ lose first mark	1	1 1		3	3	
		(ii)		If frequency is constant (1) as speed increases wavelength increases (1) Alternative [by calculation] Using 0.2 Hz (1) and calculating wavelength at a different depth (1) To award 2 marks 'suggestion not true' [or equiv] must be made			2	2		
				Question 6 total	**4**	**7**	**4**	**15**	**11**	**0**